👩🏾‍🏫 Staff Guide · Biology A · Unit 2 Algae & Photosynthesis (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 2 Algae Biofuel: A1 A look at algae (microscope), A2 Algae growth lab, A4 Pop bead photosynthesis, A4 Elodea & snails, A4.5 Stomata peel, Photosynthesis diagram, Photosynthesis video quiz, Scientific writing practice")	**Level 1 piece**
A1 A look at algae (microscope)	Slides + notes part A (microscope parts, magnification)
A2 Algae growth lab (multi-day) + scientific writing	Algae Growth Lab packet (🔬 Labs & Projects)
A4 Pop bead photosynthesis + Photosynthesis diagram	Pop Bead packet + notes part B
A4 Elodea & snails (BTB demo)	Elodea & Snails packet (observer jobs)
A4.5 Stomata peel + stomata review	Stomata Peel packet + notes part C
Photosynthesis video quiz	🚧 Not modified: the video and questions are missing
Class slides	Deck: 17 slides, Words to Know first, "point to the picture" checks, recap
Her worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	10 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Picture equation at Level 1: ☀️ + 💧 + 💨 → 🍬 + 🫧
- Level 2 uses word equations only, no chemical formulas

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part
- Cover one choice to make it 2

4. What to grade

- Unit quiz can stand in for the photosynthesis part of the Algae test
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **Photosynthesis Video Quiz**: Its questions are inside the 63 MB video that was left out of the copy; the answer sheet is just lines 1-15. Need the video or her question list.

6. Photo credits (open licenses)

Diatoms through the microscope.jpg — Prof. Gordon T. Taylor, Stony Brook University, Public domain, Wikimedia Commons · Stomata in the lower epidermis of a melon leaf with powdery mildew.jpg — Tian, M.; Yu, R.; Yang, W,; Guo, S.; Liu, S.; Du, H.; Liang, J.; Zhang, X., CC BY 4.0, Wikimedia Commons

👩🏾‍🔬 Staff Guide · Algae Growth Lab (Level 1)

Biology A · Unit 2 · goes with the class “A2 Algae Lab (+ Day 1-4 pictures)” (4 pts × 4 days). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A2 Algae lab: 3 tubes (control, dish soap, plant fertilizer) with algae and BTB; observe color, clarity, and smell for 4 days. 5 steps, one per box. **check with Ms. Dunham**: Her expected results are not in the export; our key expects fertilizer = most growth, dish soap = least
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Which tube looks the greenest?”
- “What did we add to this one?”

3. Modifications

- Circle the greenest tube; photos count as data
- Staff counts drops with the student

4. What to grade (4 pts × 4 days)

- Same points each day: photo + circle
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Diatoms through the microscope.jpg — Prof. Gordon T. Taylor, Stony Brook University, Public domain, Wikimedia Commons

👩🏾‍🔬 Staff Guide · Elodea & Snails (Observer Jobs) (Level 1)

Biology A · Unit 2 · goes with the class “A4 Elodea & snails (Day 1 + Final)” (10 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A4 Elodea & snails demo: 8 tubes of water + BTB with a snail, Elodea, both, or nothing, in light or dark for 48 hours; read the BTB color. 5 steps, one per box. **check with Ms. Dunham**: Our key gives the expected BTB colors; her real tubes may differ
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Snail: yellow or blue?”
- “Plant in the sun: yellow or blue?”

3. Modifications

- Color the 4 picture tubes
- Staff reads the tube labels

4. What to grade (10 pts)

- Same 10 points: guess + colors + check sheet
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

👩🏾‍🏫 Staff Guide · Pop Bead Photosynthesis (Level 1)

Biology A · Unit 2 · goes with the class “A4 Pop Bead Photosynthesis” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Model card**: A4 Pop bead photosynthesis: build CO2 and water from beads, split water in the light reactions, build glucose in the Calvin cycle. 4 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Show me water. Show me CO2.”
- “What does the light do?”

3. Modifications

- Build water and CO2 only; staff builds glucose
- Circle the inputs

4. What to grade (5 pts)

- Same 5 points: built + circle
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

👩🏾‍🏫 Staff Guide · Stomata Peel (Observer Jobs) (Level 1)

Biology A · Unit 2 · goes with the class “A4.5 Stomata Peel” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A4.5 Stomata peel: clear nail polish on the upper and lower leaf, tape peel onto a slide, count stomata at 400x. 4 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Point to a tiny hole.”
- “Top or bottom: which has more?”

3. Modifications

- Observer or counter only
- Staff focuses the microscope; a phone photo of the view counts

4. What to grade (5 pts)

- Same 5 points: looked + circle
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Our students only watch the nail polish step (staff paints). Please confirm the observer jobs, and which plant's leaves she uses.

6. Photo credits (open licenses)

Stomata in the lower epidermis of a melon leaf with powdery mildew.jpg — Tian, M.; Yu, R.; Yang, W,; Guo, S.; Liu, S.; Du, H.; Liang, J.; Zhang, X., CC BY 4.0, Wikimedia Commons

🌿 Algae & Photosynthesis Notes

Biology A · Unit 2 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.


algae _____


sun _____


sugar _____


oxygen _____

🖼️ B. Fill in the word. Use the pictures.

5.  Plants need the _____ to make food.

6.  Plants make _____.

7.  Plants give off _____.

8.  Leaves have tiny _____.

Word bank:  **holes**  **oxygen**  **sugar**  **sun**

⭕ C. Circle the right picture.

9. Algae live in…


Sand


Water

10. This makes tiny things look big:


Ruler


Microscope

11. BTB turns yellow with lots of…


CO2


Sugar

🔗 D. Draw a line from the picture to the word.









water

CO2

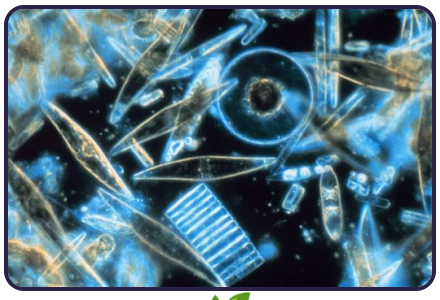
sugar

sun

📖 Algae & Photosynthesis: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Algae

tiny green water plants

Photosynthesis

plants make food from sun

Carbon dioxide

gas we breathe out

Oxygen

gas we breathe in



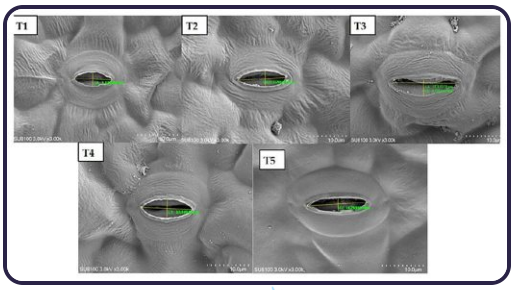
Glucose

sugar plants make



Chloroplast

green part that makes food





Stomata

tiny holes in leaves



Microscope

makes tiny things big



BTB

blue-yellow color test

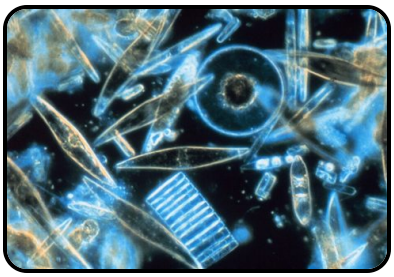


Fertilizer

plant food

🧪 Algae Growth Lab

Biology A · Unit 2 · Level 1



Name: _____

A2 Algae lab: 3 tubes (control, dish soap, plant fertilizer) with algae and BTB; observe color, clarity, and smell for 4 days · check with Ms. Dunham.

🧺 Lab stuff: get it. Check ☐.

- ☐ 🧪 **3** test tubes with caps
- ☐ 🌿 **3 droppers** algae (1 dropper per tube)
- ☐ 🔵 **5 drops each** BTB
- ☐ 🌱 **5 drops** plant fertilizer solution
- ☐ 🧼 **5 drops** dish soap solution
- ☐ 🏷️ **tape** labels + Sharpie

🌱 Do one box. Check ☐ it.

1. 🏷️ **Label** 3 tubes. ☐
2. 💧 Fill with **water**. Add **algae**. ☐
3. 🔵 Add **BTB** drops. ☐
4. 🌱 Add **soap** or **plant food**. ☐
5. 📷 **Look** each day. Take a photo. ☐

📋 Which tube is the greenest on day 4? Circle.

Control	Soap	Plant food
💧		🌱

🐌 Elodea & Snails (Observer Jobs)

Biology A · Unit 2 · Level 1



Name: _____

A4 Elodea & snails demo: 8 tubes of water + BTB with a snail, Elodea, both, or nothing, in light or dark for 48 hours; read the BTB color · check with Ms. Dunham.

🧺 Demo tubes (staff sets up): get it. Check ☐.

- ☐ 🐌 **A / E** snail (light / dark)
- ☐ 🌿 **B / F** Elodea 2 cm (light / dark)
- ☐ 🐌🌿 **C / G** both (light / dark)
- ☐ 💧 **D / H** nothing (light / dark)

🌱 Do one box. Check ☐ it.

1. 👀 **Look** at the 8 tubes. ☐
2. 🔮 **Guess**: blue or yellow? ☐
3. ⏳ **Wait** 2 days. ☐
4. 🎨 **Match** each color. ☐
5. 🐌 Snail = 🟡. Plant in light = 🔵. ☐

🎨 Color the tubes after 2 days.

🐌	🌿☀️	🌿🌙	💧

📿 Pop Bead Photosynthesis

Biology A · Unit 2 · Level 1



Name: _____

A4 Pop bead photosynthesis: build CO2 and water from beads, split water in the light reactions, build glucose in the Calvin cycle.

🧺 Bead kit: get it. Check ☐.

- ☐ ⚫ **black** carbon (C)
- ☐ 🔴 **red** oxygen (O)
- ☐ ⚪ **white** hydrogen (H)
- ☐ 🃏 **4 cards** water, atmosphere, light reactions, Calvin cycle

🌱 Do one box. Check ☐ it.

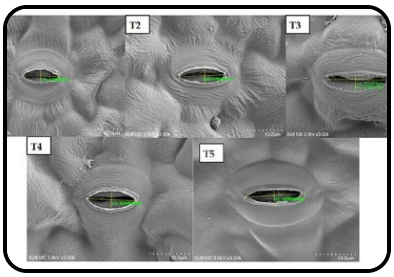
1. 📿 **Build** water: ⚪🔴⚪ ☐
2. 📿 **Build** CO2: 🔴⚫🔴 ☐
3. ☀️ **Shake** the light card. Break the water. ☐
4. 🍬 **Build** sugar and 🫧 oxygen. ☐

🔄 Circle what goes IN.

☀️ Light	💧 Water	💨 CO2	🍬 Sugar

🌿 Stomata Peel (Observer Jobs)

Biology A · Unit 2 · Level 1



Name: _____

A4.5 Stomata peel: clear nail polish on the upper and lower leaf, tape peel onto a slide, count stomata at 400x.

🧺 Lab stuff: get it. Check ☐.

- ☐ 🌿 **1** small leaf
- ☐ 💅 **clear nail polish** STAFF paints it
- ☐ 🩹 **clear tape** lifts the peel
- ☐ 🔬 **1 slide** + microscope

🌱 Do one box. Check ☐ it.

1. 👀 **Watch** staff paint the leaf. ☐
2. 🩹 **Press** tape. Peel it off. ☐
3. 🔬 **Look** in the microscope. ☐
4. 🔢 **Count** the tiny holes. ☐

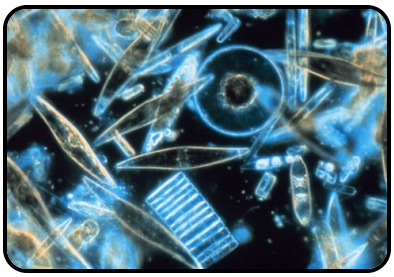
📋 Which side has more holes? Circle.

⬆️ top ⬇️ bottom

🚧 **Under construction:** Our students only watch the nail polish step (staff paints). Please confirm the observer jobs, and which plant's leaves she uses. *Check with Ms. Dunham.*

🧪 Algae Growth Lab · Get Ready

Biology A · Unit 2 · Level 1



Name: _____

👥 My job. Circle it.

🏷️	💧	📷	📝
Labeler I label.	**Dropper** I add drops.	**Photographer** I take photos.	**Recorder** I write.

🧰 Get the tools. Check ☐.

- ☐ 🏷️ Labeling tape
- ☐ 🥛 Beaker
- ☐ 📷 Camera
- ☐ 🥽 Goggles

⚠️ Be safe!

- 🥽 Goggles on.
- 👃 Waft, don't sniff.
- 🧼 Wash hands.

✅ Before we start

☐ Goggles on ☐ 3 tubes labeled ☐ Data log ready

🐌 Elodea & Snails (Observer Jobs) · Get Ready

Biology A · Unit 2 · Level 1



Name: _____

👥 My job. Circle it.

🔮	🎨	📷	📝
Predictor	**Color checker**	**Photographer**	**Recorder**
I guess.	**I match colors.**	**I take a photo.**	**I write.**

🧰 Get the tools. Check ☐.

- ☐ 🔵 BTB
- ☐ 🧪 8 tubes
- ☐ 🧻 Foil
- ☐ 🎨 Color chart

⚠️ Be safe!

- 👩‍🔬 Staff handles the tubes.
- 🥽 Goggles on.
- 🐌 Be gentle. Wash hands.

📿 Pop Bead Photosynthesis · Get Ready

Biology A · Unit 2 · Level 1



Name: _____

👥 My job. Circle it.

 **Builder** I build.	 **Breaker** I break.	 **Counter** I count.	 **Recorder** I write.

🧰 Get the tools. Check ☐.

- ☐  Pop beads
- ☐  Station cards
- ☐  This card
- ☐  Tray

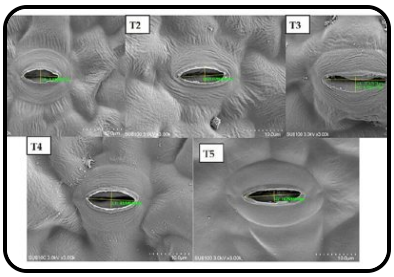
⚠️ Be safe!

 No beads in mouth.

 Pick up beads.

🍃 Stomata Peel (Observer Jobs) · Get Ready

Biology A · Unit 2 · Level 1



Name: _____

👥 My job. Circle it.

👀 Observer	🔢 Counter	🩹 Tape helper	📝 Recorder
I look.	I count.	I peel.	I write.

🧰 Get the tools. Check ☐.

- ☐  Microscope
- ☐  Slide
- ☐  Tape
- ☐  Open window/fan

⚠️ Be safe!

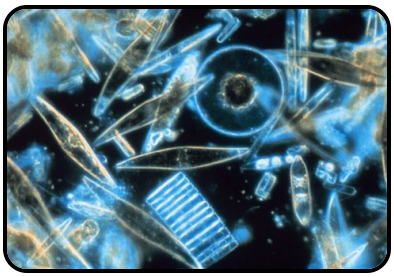
- 💅 Staff paints. I watch.
- 🔬 Two hands. Tell if glass breaks.
- 🧼 Wash hands.

✅ Before we start

☐ Staff painted the leaf (dry) ☐ Microscope on low power ☐ Data table ready

🧪 Algae Growth Lab · My Check

Biology A · Unit 2 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🧪 Day 1
- ☐ 🧪 Day 4

✅ I did it! Check ☐.

- ☐ 🏷️ I labeled
- ☐ 👀 I looked each day
- ☐ 🧠 I picked the greenest

⭕ Circle one face.

👀 **Did I check my tubes?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🐌 Elodea & Snails (Observer Jobs) · My Check

Biology A · Unit 2 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🧪 The 8 tubes

✅ I did it! Check ☐.

- ☐ 🔮 I guessed
- ☐ 🎨 I matched colors
- ☐ 🧠 I explained

⭕ Circle one face.

🎨 **Did I match the colors?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

📿 Pop Bead Photosynthesis · My Check

Biology A · Unit 2 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐  My sugar model

✅ I did it! Check ☐.

- ☐  I built molecules
- ☐  I broke water
- ☐  I counted

⭕ Circle one face.

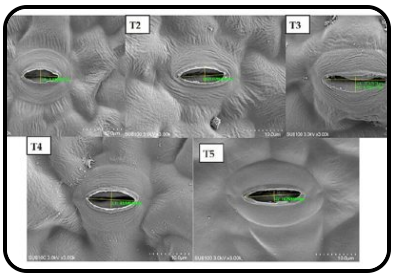
 **Did I build a model?** 😀 🙂 😕

🤝 **Did I do my job?** 😀 🙂 😕

⭐ I learned ____. Draw or write.

🍃 Stomata Peel (Observer Jobs) · My Check

Biology A · Unit 2 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🔬 My view

✅ I did it! Check ☐.

- ☐ 👀 I looked
- ☐ 🔢 I counted
- ☐ 🤝 I did my job

⭕ Circle one face.

🔬 **Did I see the holes?**	😀	🙂	😕
💅 **Was I safe?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Algae & Photosynthesis Quiz

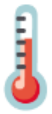
Biology A · Unit 2 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🌿 Algae live in…

-  Sand
-  Water

2. 🔬 We see tiny algae with a…

-  Microscope
-  Thermometer

3. ☀️ Plants make food with…

-  Sunlight
-  Moonlight

4. 🍬 Plants make…

-  Salt
-  Sugar

5. 🫧 Plants give off…

-  Smoke
-  Oxygen

6. Tiny holes in leaves are…

- Stomata
- Teeth

7. BTB turns yellow with lots of…

- Water
- CO2

8. Plant food that helps algae grow:

- Fertilizer
- Dish soap

🌿 Algae & Photosynthesis Notes · KEY

Biology A · Unit 2 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

 algae _____

 sun _____

 sugar _____

 oxygen _____

🖼️ B. Fill in the word. Use the pictures.

5.  Plants need the **sun** to make food.

6.  Plants make **sugar**.

7.  Plants give off **oxygen**.

8.  Leaves have tiny **holes**.

Word bank:  **holes** |  **oxygen** |  **sugar** |  **sun**

⭕ C. Circle the right picture.

9. Algae live in…

-  Sand
-  **Water** ✅

10. This makes tiny things look big:

-  Ruler
-  **Microscope** ✅

11. BTB turns yellow with lots of…

-  **CO2** ✅
-  Sugar

🔗 D. Draw a line from the picture to the word.

Key: ☀️ → sun · 💧 → water · 💨 → CO2 · 🍬 → sugar

	**water**
	**CO2**
	**sugar**
	**sun**

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Algae & Photosynthesis Quiz · KEY

Biology A · Unit 2 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🌿 Algae live in… A1 Look at algae

- Sand
- **Water** ✓

2. 🔬 We see tiny algae with a… A1 Look at algae

- **Microscope** ✓
- Thermometer

3. ☀️ Plants make food with… A4 Pop beads

- **Sunlight** ✓
- Moonlight

4. 🍬 Plants make… A4 Pop beads

- Salt
- **Sugar** ✓

5. 🫧 Plants give off… A4 Pop beads

- Smoke
- **Oxygen** ✓

6. Tiny holes in leaves are… A4.5 Stomata

- **Stomata** ✓
- Teeth

7. BTB turns yellow with lots of… A4 Elodea & snails

- Water
- **CO2** ✓

8. Plant food that helps algae grow: A2 Algae lab

- **Fertilizer** ✓
- Dish soap

Answers

1-B (Water) · 2-A (Microscope) · 3-A (Sunlight) · 4-B (Sugar) · 5-B (Oxygen) · 6-A (Stomata) · 7-B (CO2) · 8-A (Fertilizer)

Level 1 version of the gen ed "A1 Look at algae, A2 Algae lab, A4 Pop beads, A4 Elodea & snails, A4.5 Stomata". Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from her A1-A4.5 sheets, please check. Her Photosynthesis Video Quiz could not be modified (the video and questions are missing).*